

Karina Hwang Arcolezi, Ph.D.

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EDUCATION

Ph.D. in Construction Engineering, École de technologie supérieure (ÉTS), Montréal, QC, 2022–2026

Thesis title: *Design and performance of fiber-optic light-transmitting cementitious composites* [Link]

M.Sc. in Civil Engineering, São Paulo State University (UNESP), Brazil, 2020–2022 Thesis: *Influence of aggregate sizes and packing combination on the properties of pervious concrete* [Link]

B.Sc. in Civil Engineering, State University of Mato Grosso (UNEMAT), Brazil, 2014–2019

PUBLICATIONS

Journal Articles

1. **Arcolezi, K. H.**, Zimmermann, C. A., Ung, B., & Ouellet-Plamondon, C. (2026). Experimental and theoretical investigation of optical transmittance in light-transmitting concrete. *Measurement Science and Technology*, 37, 195201. <https://doi.org/10.1088/1361-6501/ae6521>
2. **Arcolezi, K. H.**, Marion, V., Ung, B., & Ouellet-Plamondon, C. (2025). Randomly distributed optical fibers in translucent mortar for privacy-preserving light transmission and digital image reconstruction. *Scientific Reports*, 16, 2454. <https://doi.org/10.1038/s41598-025-32224-2>
3. da Silva, R. G., **Arcolezi, K. H.**, Lyra, G. P., Akasaki, J. L., Soriano, L., Borrachero, M. V., Payá, J., & Tashima, M. M. (2025). New insights on the development of sustainable pervious concrete using sand replacing paste volume – Technical, economical, and environmental analysis. *Journal of Building Engineering*, 116, 114657. <https://doi.org/10.1016/j.jobbe.2025.114657>
4. **Arcolezi, K. H.**, da Silva, R. G., Soriano, L., Borrachero, M. V., Monzó, J., Payá, J., Tashima, M. M., & Akasaki, J. L. (2023). Effect of aggregates packing with the maximum density methodology in pervious concrete. *Sustainability*, 15(6), 4939. <https://doi.org/10.3390/su15064939>

Submitted Manuscripts

5. **Arcolezi, K. H.**, Brial, V., Ung, B., & Ouellet-Plamondon, C. (2026). Mechanical strengthening of cementitious composites by optical fibers through adhesion and geometric disruption. Manuscript submitted for publication.

Conference Proceedings

6. **Arcolezi, K. H.**, Alonso Romero, A., Ben Zaid, H., Hamel, T., Ung, B., & Ouellet-Plamondon, C. (2026). Touch-interactive concrete for smart structures. *Proceedings*

of the Canadian Society for Civil Engineering Annual Conference 2026. Accepted, to appear.

7. **Arcolezi, K. H.**, Ouellet-Plamondon, C., & Ung, B. (2026). Influence of optical fibres on mechanical properties of mortar mixes. In A. H. M. M. Billah & A. El Ansary (Eds.), *Proceedings of the Canadian Society for Civil Engineering Annual Conference 2024, Volume 4* (Lecture Notes in Civil Engineering, vol. 717). Springer, Cham. https://doi.org/10.1007/978-3-031-98775-5_23
8. **Arcolezi, K. H.**, Ouellet-Plamondon, C., & Ung, B. (2025). Interactive light- and sound-sensitive mortar for architectural environments. In A. Francis, E. Miresco, & S. Melhado (Eds.), *Advances in Information Technology in Civil and Building Engineering. ICCCB E 2024* (Lecture Notes in Civil Engineering, vol. 629). Springer, Cham. https://doi.org/10.1007/978-3-031-87364-5_41

PRESENTATIONS

Oral Presentations

- *Touch-interactive concrete for smart structures*. Canadian Society for Civil Engineering (CSCE) Annual Conference, June 2026.
- *Interactive light- and sound-sensitive mortar for architectural environments*. International Conference on Computing in Civil and Building Engineering (ICCCBE), Montréal, 2024.
- *Influence of optical fibres on the mechanical properties of mortar mixes*. Canadian Society for Civil Engineering (CSCE) Annual Conference, 2024.

Poster Presentations

- *Optical fiber-embedded smart concrete for sensing and illumination*. 36th Annual Conference of the Centre for Optics, Photonics and Lasers (COPL), 2025.

Art-Science Exhibitions

- *From Quantum Science to Artistic Experience*, INTRIGO Collective, Amos Exhibition Centre, Amos, QC, 2025. Light-transmitting cementitious composites presented as immersive installations exploring the translation of quantum sciences into sensory experiences [Link]

RESEARCH AND PROFESSIONAL EXPERIENCE

Ph.D. Researcher, Construction Engineering, ÉTS, Montréal, QC, 2022–2026 Led a doctoral project on optical fiber-embedded cementitious composites, combining low-carbon material design, optical characterization, mechanical testing, privacy-preserving light transmission, digital image reconstruction, and interactive sensing. Collaborated with researchers in construction materials, optics, sensing, and smart systems.

Research Intern, University of Sydney, Sydney, Australia, Dec. 2021–Feb. 2022 Contributed to a flood risk assessment project in Indonesia using GIS-based spatial analysis (QGIS, JOSM), covering infrastructure exposure, climate-related flooding, and sea-level rise.

Teaching Assistant, Building Construction, São Paulo State University, Brazil, Mar.–Sept. 2021 Supported a 60-hour undergraduate course in building construction.

Civil Engineering Intern, Municipality of Tangará da Serra, Brazil, 2018 Supported engineering documentation and public-sector construction activities in the municipal infrastructure department.

FUNDING, AWARDS AND DISTINCTIONS

Scholarships and Fellowships

- Fonds de recherche du Québec – Nature et technologies (FRQNT), Merit Scholarship Program for Foreign Students (PBEEE), 2025–2026.
- École de technologie supérieure, Doctoral Scholarship, 2022–2026.
- University of Sydney, Summer Research Scholarship, 2021–2022.

Research Support

- École de technologie supérieure, Financial support for the dissemination and promotion of research work, 2024 and 2026.

Awards and Distinctions

- Acfas, *La preuve par l'image* research photo contest. Image selected among the 20 featured works, exhibited at the Biodôme de Montréal, 2026 [Link]
- École de technologie supérieure, Photo Contest, Arts and Culture category, Winner, 2025.
- NSERC, *Science Exposed* research photo contest, Jury Prize, 2024 [Link]

SKILLS

Research: experimental design, optical characterization, mechanical testing, image processing, optoelectronic systems, smart materials, GIS-based environmental risk analysis, data analysis and visualization, scientific writing

Tools: Python, Excel, QGIS, JOSM, LaTeX, Inkscape, Canva, Microsoft Office

Languages: French, English, Portuguese